

COSC 1100



Class Exercise



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Planning: Your First Programming Assignment
Part 1 of 2



Planning: Your First Programming Assignment

Goal of Part 1:

Experience working with the initial steps of the simple planning process that was introduced in the videos, with a focus on fully understanding the problem.

Requirements

- Did you come prepared?
 - Getting Started in COSC 1100 videos, ideally all done!
 - Planning an Application videos, ideally at least started!
- Collaboration.
 - You will be encouraged to work together with 1-2 colleagues during this exercise. (i.e. groups of 2-3)
 - Students currently outside of Canada are encouraged to communicate and collaborate via MS Teams.
 - The group will share their planning ideas at the end of this class, which will contribute to your grade.

What Is An Assignment?

- Seriously, what is it?
- Why do we do it?
- Is an assignment a good thing or a bad thing?

A Simple Programming Approach (Steps 1 – 10)

1. Clearly define the result (**OUTPUT**)
2. Determine what data is needed, and how to get it (**INPUT**)
3. List the steps needed in between (**PROCESS**)
4. Write your **PLAN** as pseudo-code and/or a flowchart
5. Test your plan for logical errors (**DESK-CHECK**)
6. Get your development environment ready (**SET UP**)
7. Translate your plan into code (**CODE**)
8. Attempt to run your program; find and fix syntax errors (**DEBUG**)
9. Run your program using test data (**TEST**)
10. Distribute

Step 1: Output (Review)

- Clearly define the result (**OUTPUT**)
 - Commission for a salesperson
 - Dollar amount
 - Floating-point variable
 - Formatted as currency
 - Embedded in a string, i.e. "This salesperson's commission is "
 - Should we also show the commission rate?

Understand the Problem

- Read the following problem statement, *for detailed comprehension*.
 1. Analyze the problem requirements and put your notes in a useful, sharable format.

In other words, complete steps 1, 2 and 3 of the planning approach explained in the videos.
 2. Be prepared to discuss your notes in class.
 3. Make a discussion post including your notes on DC Connect (**COSC 1100 -> Activities -> Discussion -> Class Exercises**), including all group members' names.

The Problem

A small, local ice cream shop is trying to conservatively forecast their stock of ice cream. They sell kiddie cones (1/2 scoop), small cones (1 scoop), medium cones (2 scoops) and sometimes large ones (3 scoops). A scoop is generally measured as roughly $\frac{1}{2}$ cup of ice cream, or 120 mL.

To maximize expiration dates and minimize cold storage, given a certain amount of each size that was sold, they want to identify the amount of ice cream sold in 1 week.

Class Exercise Part 1 Discussion

Consider the notes presented and things other students have said.

- Were they much different than your group?
- In what ways were their notes helpful compared to trying to write a plan or code directly from the problem description?
- What constructive ways could their notes be improved?

Class Exercise Part 1 Summary

- Compared to the other steps of the planning approach we have been discussing (4 - 10)... how important is it that you **understand the problem** (steps 1 - 3)?
- Be ready to discuss the next few steps very shortly.

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Planning: Your First Programming Assignment
Part 2 of 2



Planning: Your First Programming Assignment

Goal of Part 2:

Experience working toward a complete, working program starting from a mostly-complete plan as described in the course's video content thus far.

Requirements

- Did you come prepared?
 - Getting Started in COSC 1100 videos, ideally all done!
 - Planning an Application videos, ideally these are done too!
- Collaboration.
 - You may continue to work together with 1-2 colleagues during this exercise. (i.e. groups of 2-3)
 - Students currently outside of Canada are encouraged to communicate and collaborate via MS Teams.
 - Each group will submit their finished application for marks.

The Problem (Review)

A small, local ice cream shop is trying to conservatively forecast their stock of ice cream. They sell kiddie cones (1/2 scoop), small cones (1 scoop), medium cones (2 scoops) and sometimes large ones (3 scoops). A scoop is generally measured as roughly $\frac{1}{2}$ cup of ice cream, or 120 mL.

To maximize expiration dates and minimize cold storage, given a certain amount of each size that was sold, they want to identify the amount of ice cream sold in 1 week.

Formalizing the Plan

- Step 1 – 3, simplified, are describing the Output, the Input, and the Processing, in that order.
- Step 4 is organizing and formalizing those steps, into pseudocode or a diagram (like a flowchart).
- Let's take the best parts of the Steps 1 – 3 we worked on in Part 1, and do Step 4 together.
- Then we'll have a real plan!

Plan to Program!

- With consideration to the plan we just developed, code the application using Python!
 1. Code your solution to the given problem based on the plan provided or your own planning notes.
 2. Chat, ask lots of questions, and don't be afraid of mistakes.
 3. Either have it checked during class,
or submit your group's finished Python code file to DC Connect
(**COSC 1100 -> Activities -> Assignments -> Class Exercise: First Python Program**), including all group members' names.

This is meant to be done during class time; submissions received after our class time ends may be considered late.

Class Exercise Part 2 Discussion

Consider the process of creating a simple Python application.

- We skipped one step of the planning process! Did anyone notice? What did we miss?
- What, if any, is the benefit of working and consulting with other students?

Class Exercise Part 2 Summary

- You've created and submitted your first program for this course!
- Is it more important to develop good programming habits than it is to be skillful with coding syntax? Why or why not?